

AI Powered Used Car Price Prediction Using Machine Learning

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Internship : Data Science Internship

Organization : iNeuBytes

Submission Date : August 2026

Abstract:

The rapid growth of the used car market has made it difficult for buyers and sellers to determine the fair value of a vehicle. This project presents an AI-powered Used Car Price Prediction system that estimates the market value of a used car using Machine Learning techniques. The system considers several vehicle attributes such as brand, model, manufacturing year, mileage, fuel type, transmission, engine specifications, exterior and interior colors, accident history, and title status. A Random Forest Regressor model is trained on a preprocessed dataset to provide accurate price predictions. The application is deployed using Streamlit, allowing users to enter vehicle details through a user-friendly interface and receive instant predictions.

Introduction:

The automobile industry has experienced significant growth over the past decade, leading to an increasing demand for used vehicles. Determining the correct resale price is often challenging because many factors influence the value of a vehicle. Buyers frequently overpay, while sellers may undervalue their vehicles due to a lack of market knowledge.

Machine Learning offers an efficient solution by learning patterns from historical vehicle sales data and predicting prices based on vehicle specifications. This project aims to simplify the pricing process by providing accurate and reliable price predictions through an interactive web application.

Problem Statement:

Estimating the resale value of used vehicles manually is time-consuming and often inaccurate. Vehicle prices depend on numerous factors such as mileage, brand reputation, fuel type, model year, accident history, and engine specifications. Traditional pricing methods rely heavily on human judgment and market experience.

The objective of this project is to develop an intelligent system capable of predicting used car prices accurately using Machine Learning algorithms.

Objectives:

The main objectives of this project are:

- To analyze a used car dataset.
- To preprocess and clean the collected data.
- To perform feature engineering for better model performance.
- To train a Machine Learning model for price prediction.
- To evaluate the model using standard performance metrics.
- To deploy the model through a Streamlit web application.
- To provide users with accurate market price predictions.

Dataset Description:

The project uses a publicly available Used Car dataset containing information about thousands of vehicles.

Dataset Features

- Brand
- Model
- Model Year
- Mileage

- Fuel Type
- Engine Specification
- Transmission
- Exterior Color
- Interior Color
- Accident History
- Clean Title
- Selling Price

Dataset Statistics

- Total Records: **4009**
- Total Brands: **57**
- Average Price: **₹37,42,468**
- Maximum Price: **₹24,81,42,972**

The dataset contains both numerical and categorical attributes, making it suitable for supervised regression problems.

Data Preprocessing:

The collected dataset was cleaned and prepared before training the Machine Learning model.

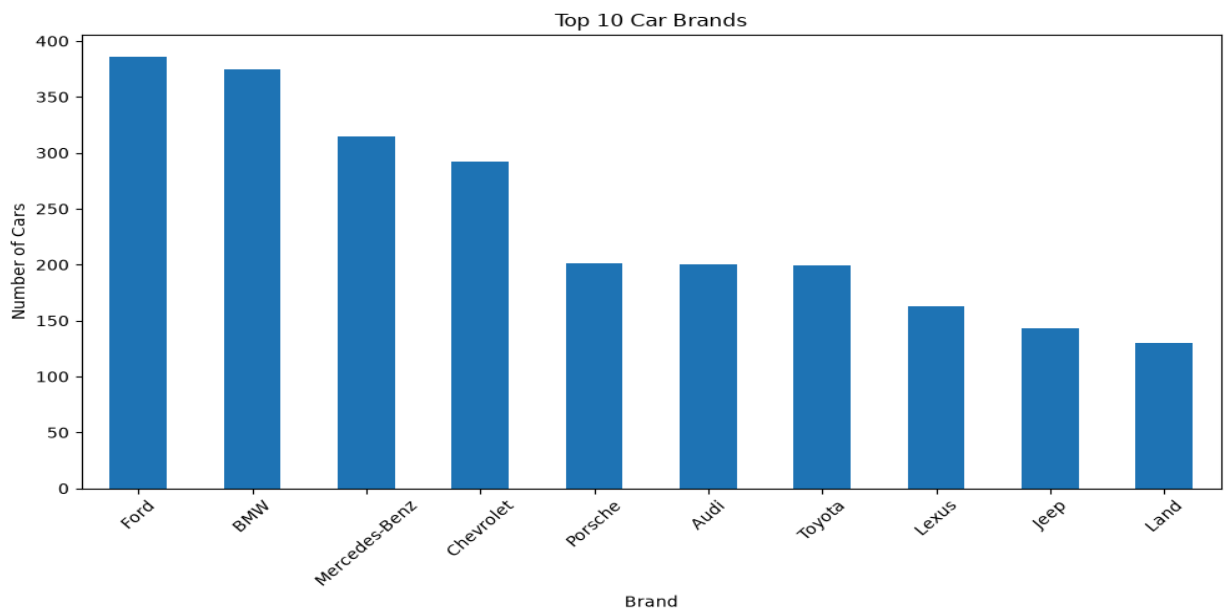
The preprocessing steps included:

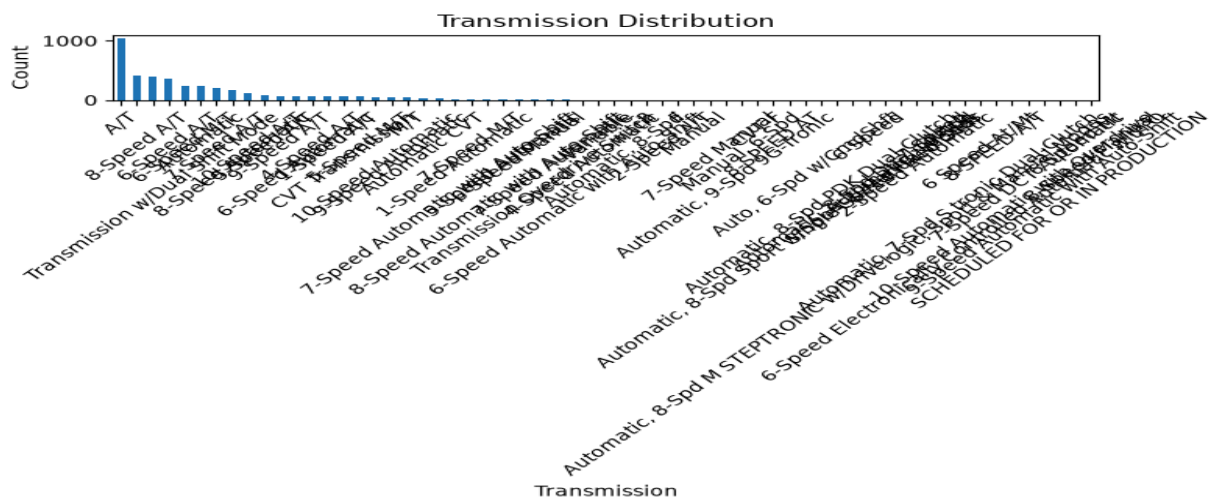
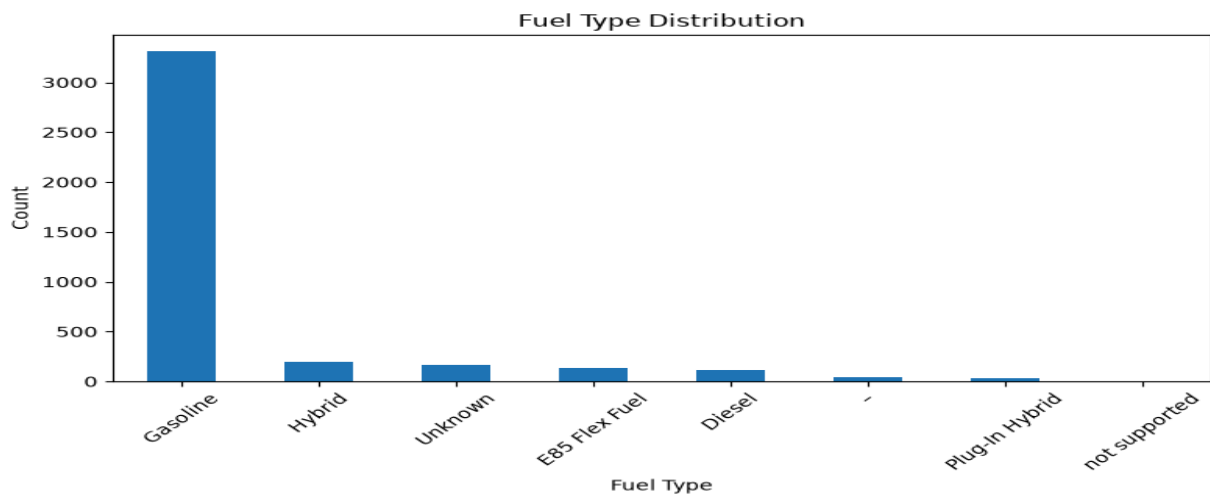
- Removing duplicate records
- Handling missing values
- Correcting inconsistent data
- Encoding categorical variables
- Converting numerical columns into appropriate data types
- Feature selection
- Splitting the dataset into training and testing sets

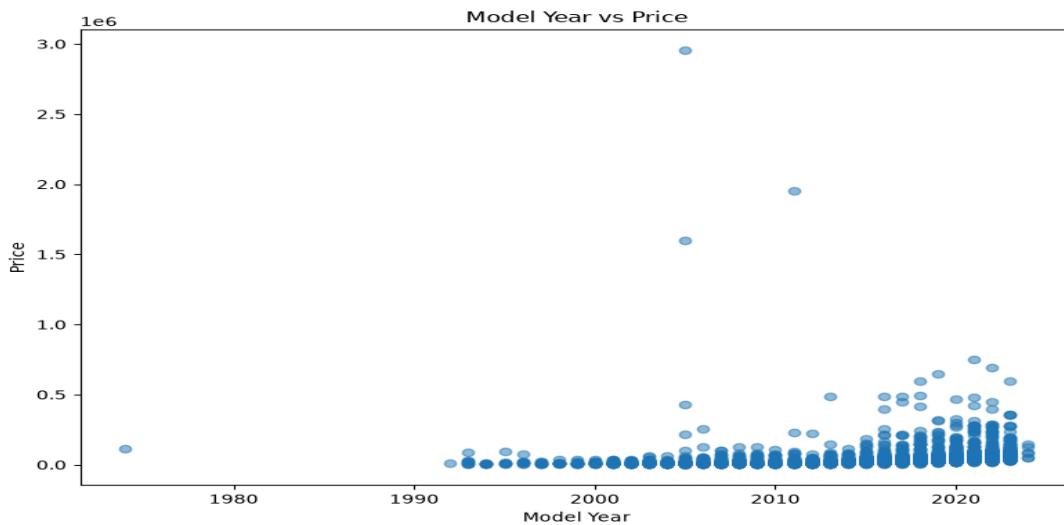
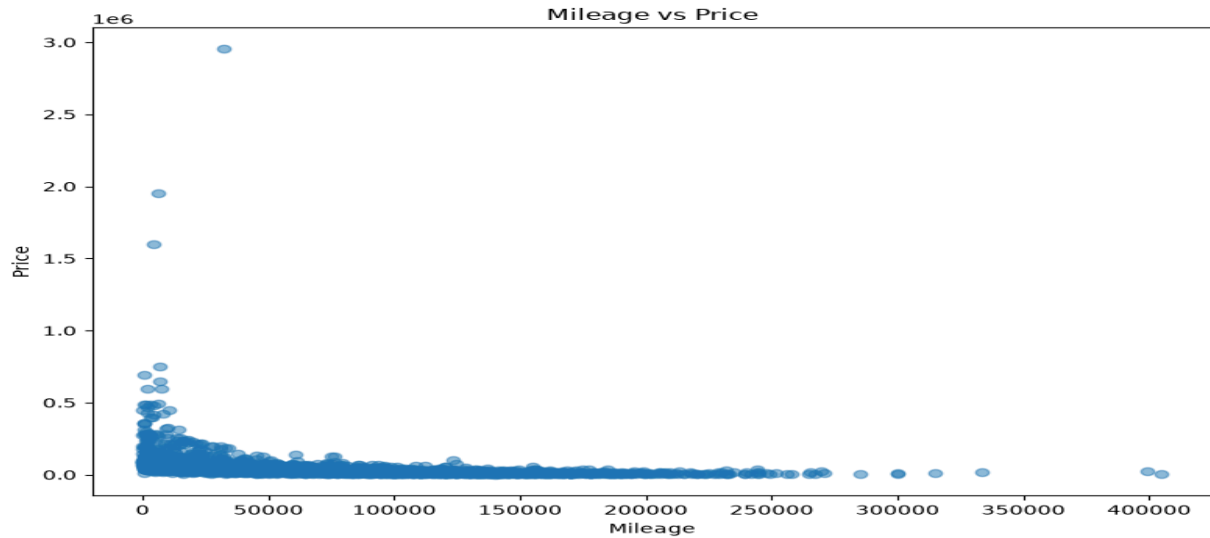
These preprocessing techniques improved the quality of the dataset and enhanced prediction accuracy.

Exploratory Data Analysis (EDA):

Exploratory Data Analysis (EDA) was performed to understand the structure of the dataset, identify trends, detect missing values, and analyze relationships between different vehicle features before training the Machine Learning model.







Feature Engineering:

Feature engineering plays an important role in improving Machine Learning performance.

The following features were selected:

- Brand
- Model
- Fuel Type

- Engine
- Transmission
- Exterior Color
- Interior Color
- Model Year
- Mileage
- Accident History
- Clean Title

Categorical variables were encoded before training the model.

Machine Learning Model Development:

Several regression algorithms were considered during the development process. After experimentation, the Random Forest Regressor was selected because it produced the most accurate predictions.

Why Random Forest?

- High prediction accuracy
- Handles nonlinear relationships effectively
- Reduces overfitting
- Works well with mixed numerical and categorical data
- Requires minimal parameter tuning

The model was trained using the preprocessed dataset and validated using train-test splitting.

Model Training:

The dataset was divided into training and testing datasets.

- Training Data: 80%
- Testing Data: 20%

The Random Forest model learned the relationship between vehicle features and selling prices during training.

Model Evaluation:

The trained model was evaluated using regression performance metrics.

Evaluation Metrics

- R^2 Score
- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)

These metrics help measure how closely the predicted prices match the actual vehicle prices.

The Random Forest model demonstrated reliable prediction performance with low prediction error and strong generalization capability.

Deployment:

The trained Machine Learning model was deployed using **Streamlit**.

The application provides a simple and interactive interface where users can enter vehicle details and receive an estimated market value instantly.

Features of the Application:

- Home Page
- Price Prediction Module
- Dashboard
- Model Performance Page
- About Section

User Interface:

The application provides an easy-to-use interface for users.

Users enter:

- Brand
- Model
- Fuel Type
- Engine
- Transmission
- Exterior Color
- Interior Color
- Accident History
- Clean Title
- Model Year
- Mileage

After clicking the **Predict Price** button, the application displays:

- Estimated Market Value
- Acceptable Price Range
- Selected Vehicle Details

Technologies Used:

Technology	Purpose
Python	Programming Language
Pandas	Data Processing

NumPy	Numerical Computation
Scikit-learn	Machine Learning
Random Forest Regressor	Prediction Model
Streamlit	Web Application
Matplotlib	Data Visualization
Plotly	Interactive Charts
Joblib	Model Saving

Testing:

The deployed application was tested using different vehicle combinations.

Testing confirmed that:

- User inputs were accepted correctly.
- Predictions were generated successfully.
- The application responded quickly.
- Vehicle information was displayed correctly.
- The estimated price varied according to user inputs.

The system performed reliably under different test conditions.

Results:

The developed system successfully predicts used car prices using Machine Learning.

Achievements

- Accurate price prediction
- User-friendly interface
- Fast prediction time
- Interactive dashboard
- Reliable performance
- Easy deployment using Streamlit

The application provides estimated market values along with an acceptable demand range, helping buyers and sellers make informed decisions.

Advantages:

- Reduces manual price estimation.
- Saves time.
- Improves pricing accuracy.
- Easy to use.
- Interactive user interface.
- Supports multiple vehicle attributes.
- Helps buyers and sellers make better decisions.

Future Scope:

Future improvements may include:

- Live market price integration
- Vehicle image upload
- VIN number-based prediction
- Deep Learning models
- Mobile application

- Cloud deployment
- Real-time dealer price comparison
- AI-powered recommendation system

Conclusion:

This project successfully demonstrates how Machine Learning can be used to predict used car prices accurately. The Random Forest Regressor effectively learns patterns from historical vehicle data and generates reliable market value predictions. The Streamlit application provides a simple and interactive platform that enables users to obtain instant price estimates by entering vehicle details. Overall, the project highlights the practical application of Machine Learning in solving real-world pricing problems while offering a scalable solution for buyers, sellers, and automobile businesses.

GitHub Repository

Repository Name:

AI-Used-Car-Price-Prediction

Repository Contents

- Source Code
- Dataset
- Machine Learning Model
- Streamlit Application
- README File

- Requirements File

Screenshots:

The screenshot displays a web application interface for "AI Powered Used Car Market Analytics". On the left is a sidebar with a red car icon and the title "AI Used Car". Below this is a "Navigation" menu with five items: "Home" (selected), "Price Prediction", "Dashboard", "Model Performance", and "About". The main content area features a title "AI Powered Used Car Market Analytics" with a car icon, a welcome message, and a description of the application's purpose. A "Features" section lists five capabilities: AI Price Prediction, Interactive Dashboard, Model Performance, Market Analytics, and User Friendly Interface. Below this is a table with four columns: "Total Cars", "Brands", "Average Price (INR)", and "Maximum Price (INR)". The table contains the following data: Total Cars: 4009, Brands: 57, Average Price (INR): ₹37,42,468, and Maximum Price (INR): ₹24,81,42,972. At the bottom of the main content area is a section titled "Dataset Preview". A "Deploy" button is located in the top right corner of the interface.

Total Cars	Brands	Average Price (INR)	Maximum Price (INR)
4009	57	₹37,42,468	₹24,81,42,972



AI Used Car

Navigation

- Home
- Price Prediction**
- Dashboard
- Model Performance
- About

Deploy

AI Used Car Price Prediction

Fill in the vehicle details below.

Brand

Audi

Exterior Color

White

Model

A5 45 S line Premium Plus

Interior Color

Black

Fuel Type

Hybrid

Accident History

None reported

Engine

261.0HP 2.0L 4 Cylinder Engine Gasoline/Mild Electric Hybrid

Clean Title

Yes

Transmission

Transmission w/Dual Shift Mode

Model Year

2022

Mileage

25000



AI Used Car

Navigation

- Home
- Price Prediction**
- Dashboard
- Model Performance
- About

Deploy

Predict Price

ESTIMATED MARKET VALUE

₹42,32,684

Acceptable Demand Range ($\pm 7\%$)

₹39,36,396 - ₹45,28,972

Model: Audi A5 45 S line Premium Plus (2022)

Mileage: 25,000 mi | Fuel: Hybrid

